

Brutal Practice Paper B34 - Reproduction in Plants, Lines & Angles, Wastewater Story, Comparing Quantities

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A sunflower head holds 1200 seeds, but 35 percent of them fail to sprout. How many of its seeds actually germinate? [\[Reproduction in Plants\]](#)
 - (A) 1135 seeds
 - (B) 800 seeds
 - (C) 65 seeds
 - (D) 1165 seeds
 - (E) 780 seeds
 - (F) 765 seeds

2. Two complementary angles are in the ratio 2 : 3. What is the measure of the larger angle? [\[Lines & Angles\]](#)
 - (A) 30 degrees
 - (B) 45 degrees
 - (C) 54 degrees
 - (D) 36 degrees
 - (E) 60 degrees
 - (F) 27 degrees
 - (G) 72 degrees

3. A city produces 360 million litres of sewage each day, and its treatment plant can handle 75 percent of it. How many million litres are treated each day? [\[Wastewater Story\]](#)
 - (A) 135 million L
 - (B) 300 million L
 - (C) 90 million L
 - (D) 270 million L
 - (E) 360 million L
 - (F) 180 million L

4. A shopkeeper buys a bag for Rs 800 and sells it at a 25 percent profit, then gives a loyal customer a further 10 percent off that selling price. How much does the customer finally pay? [\[Comparing Quantities\]](#)
 - (A) Rs 1000
 - (B) Rs 950
 - (C) Rs 990
 - (D) Rs 900
 - (E) Rs 720
 - (F) Rs 1100
 - (G) Rs 810

5. A farmer plants one onion bulb. It produces 6 bulblets, and the next year each of those bulblets produces 6 more. How many bulblets are formed in the second year? [\[Reproduction in Plants\]](#)
- (A) 66 bulblets
 - (B) 216 bulblets
 - (C) 12 bulblets
 - (D) 30 bulblets
 - (E) 36 bulblets
 - (F) 42 bulblets
 - (G) 6 bulblets
6. An angle is three times the size of its own supplement. What is the measure of the angle? [\[Lines & Angles\]](#)
- (A) 108 degrees
 - (B) 120 degrees
 - (C) 135 degrees
 - (D) 60 degrees
 - (E) 150 degrees
 - (F) 45 degrees
 - (G) 90 degrees
7. A leaking tap drips 3 drops every second, and 20 drops make 1 millilitre of water. How many millilitres of water are wasted in one hour? [\[Wastewater Story\]](#)
- (A) 180 mL
 - (B) 10800 mL
 - (C) 540 mL
 - (D) 5400 mL
 - (E) 108 mL
 - (F) 1080 mL
8. A shirt marked at Rs 1200 is sold at a 30 percent discount, after which a 5 percent tax is added to the discounted price. What is the final amount paid? [\[Comparing Quantities\]](#)
- (A) Rs 924
 - (B) Rs 900
 - (C) Rs 840
 - (D) Rs 810
 - (E) Rs 800
 - (F) Rs 882
 - (G) Rs 888
9. A flower has 6 petals, 6 sepals, 12 stamens and 1 pistil. What percentage of all these flower parts are stamens? [\[Reproduction in Plants\]](#)
- (A) 48%
 - (B) 52%
 - (C) 24%
 - (D) 40%
 - (E) 50%
 - (F) 12%

- 10.** Two angles form a linear pair, and one is 24 degrees more than the other. What is the measure of the larger angle? [\[Lines & Angles\]](#)
- (A) 72 degrees
 - (B) 102 degrees
 - (C) 108 degrees
 - (D) 78 degrees
 - (E) 66 degrees
 - (F) 90 degrees
 - (G) 114 degrees
- 11.** A household uses 1200 litres of water a day. After fixing its leaks, water use drops by 15 percent. How many litres does the household now use each day? [\[Wastewater Story\]](#)
- (A) 1015 L
 - (B) 900 L
 - (C) 1185 L
 - (D) 180 L
 - (E) 1380 L
 - (F) 1020 L
 - (G) 1170 L
- 12.** The price of rice rose from Rs 40 per kilogram to Rs 50 per kilogram. What is the percentage increase in price? [\[Comparing Quantities\]](#)
- (A) 50%
 - (B) 25%
 - (C) 10%
 - (D) 20%
 - (E) 12.5%
 - (F) 80%
- 13.** A dandelion plant releases 200 fluffy seeds. About 12 percent of them land on fertile soil, and of those only 25 percent germinate. How many seeds finally germinate? [\[Reproduction in Plants\]](#)
- (A) 50 seeds
 - (B) 24 seeds
 - (C) 18 seeds
 - (D) 6 seeds
 - (E) 48 seeds
 - (F) 30 seeds
- 14.** Two vertically opposite angles are given as $(2x + 10)$ degrees and 70 degrees. What is the value of x ? [\[Lines & Angles\]](#)
- (A) 35
 - (B) 25
 - (C) 30
 - (D) 15
 - (E) 40
 - (F) 20
 - (G) 60

15. Sewage is about 99.9 percent water by mass. In 5000 kilograms of sewage, what is the mass of the impurities? [\[Wastewater Story\]](#)

- (A) 50 kg
- (B) 500 kg
- (C) 5 kg
- (D) 2.5 kg
- (E) 10 kg
- (F) 0.5 kg
- (G) 4995 kg

16. A number is first increased by 20 percent and then the result is decreased by 20 percent, giving 96. What was the original number? [\[Comparing Quantities\]](#)

- (A) 100
- (B) 90
- (C) 80
- (D) 112
- (E) 104
- (F) 120

17. A maize tassel makes about 25000 pollen grains, but only 0.2 percent of them ever reach an egg cell. How many pollen grains reach an egg? [\[Reproduction in Plants\]](#)

- (A) 500 grains
- (B) 5 grains
- (C) 250 grains
- (D) 25 grains
- (E) 5000 grains
- (F) 50 grains
- (G) 50000 grains

18. Three angles meeting at a single point are in the ratio 2 : 3 : 5. What is the measure of the largest angle? [\[Lines & Angles\]](#)

- (A) 150 degrees
- (B) 72 degrees
- (C) 90 degrees
- (D) 144 degrees
- (E) 180 degrees
- (F) 120 degrees
- (G) 108 degrees

19. A sewer pipe carries 250 litres of wastewater every minute. How many kilolitres of wastewater pass through it in 2 hours? [\[Wastewater Story\]](#)

- (A) 30 kL
- (B) 300 kL
- (C) 3 kL
- (D) 60 kL
- (E) 15 kL
- (F) 500 kL
- (G) 45 kL

20. Two numbers are in the ratio 5 : 7, and their sum is 96. What is the larger of the two numbers? [\[Comparing](#)

[Quantities\]](#)

- (A) 60
- (B) 63
- (C) 49
- (D) 42
- (E) 48
- (F) 40
- (G) 56

21. Three pea pods are opened and found to hold 6, 8 and 10 seeds. What is the average number of seeds per pod? [\[Reproduction in Plants\]](#)

- (A) 12 seeds
- (B) 6 seeds
- (C) 9 seeds
- (D) 7 seeds
- (E) 10 seeds
- (F) 8 seeds

22. A pair of co-interior (allied) angles lies between two parallel lines, and one of them measures 110 degrees. What is the measure of the other? [\[Lines & Angles\]](#)

- (A) 80 degrees
- (B) 110 degrees
- (C) 90 degrees
- (D) 140 degrees
- (E) 100 degrees
- (F) 70 degrees
- (G) 55 degrees

23. At a treatment plant, 8 aeration tanks each clean 4500 litres of wastewater every hour. How many litres do all the tanks together clean in one hour? [\[Wastewater Story\]](#)

- (A) 4508 L
- (B) 72000 L
- (C) 45000 L
- (D) 36000 L
- (E) 18000 L
- (F) 3600 L
- (G) 360000 L

24. A student scored 360 marks out of a possible 500. What percentage did the student score? [\[Comparing](#)

[Quantities\]](#)

- (A) 68%
- (B) 70%
- (C) 36%
- (D) 75%
- (E) 140%
- (F) 72%
- (G) 80%

25. A rose cutting grows roots in 14 days, while a money-plant cutting takes 21 days. The rose roots faster by a number of days equal to what percentage of the money-plant's time? [\[Reproduction in Plants\]](#)

- (A) 25%
- (B) 30%
- (C) 35%
- (D) 66.7%
- (E) 7%
- (F) 33.3%

26. An angle is 14 degrees less than its own complement. What is the measure of the angle? [\[Lines & Angles\]](#)

- (A) 52 degrees
- (B) 76 degrees
- (C) 32 degrees
- (D) 45 degrees
- (E) 44 degrees
- (F) 26 degrees
- (G) 38 degrees

27. A treatment process removes 90 percent of the solids in the first stage, then removes 60 percent of whatever solids remain in the second stage. What percentage of the original solids is still left at the end?

[\[Wastewater Story\]](#)

- (A) 40%
- (B) 10%
- (C) 16%
- (D) 30%
- (E) 36%
- (F) 6%
- (G) 4%

28. What is the simple interest earned on Rs 5000 kept at 8 percent per year for 3 years? [\[Comparing Quantities\]](#)

- (A) Rs 1300
- (B) Rs 800
- (C) Rs 1500
- (D) Rs 120
- (E) Rs 400
- (F) Rs 1200
- (G) Rs 3600

29. In yeast, one cell buds into 2 cells every 20 minutes. Starting from a single yeast cell, how many cells are there after one full hour? [\[Reproduction in Plants\]](#)

- (A) 60 cells
- (B) 4 cells
- (C) 8 cells
- (D) 3 cells
- (E) 6 cells
- (F) 16 cells
- (G) 2 cells

30. An angle is exactly twice the size of its own complement. What is the measure of the angle? [[Lines & Angles](#)]

- (A) 45 degrees
- (B) 90 degrees
- (C) 60 degrees
- (D) 50 degrees
- (E) 40 degrees
- (F) 120 degrees

31. Drinking water is disinfected with 2 milligrams of chlorine per litre. How many grams of chlorine are needed to treat 50000 litres of water? [[Wastewater Story](#)]

- (A) 100000 g
- (B) 25 g
- (C) 10 g
- (D) 50 g
- (E) 200 g
- (F) 1000 g
- (G) 100 g

32. A sum of money doubles itself in 10 years under simple interest. What is the rate of interest per year? [[Comparing Quantities](#)]

- (A) 20%
- (B) 8%
- (C) 5%
- (D) 1%
- (E) 2%
- (F) 10%

33. A fern releases 5000 spores, but 96 percent of them die before growing. How many spores survive? [[Reproduction in Plants](#)]

- (A) 50 spores
- (B) 20 spores
- (C) 480 spores
- (D) 4800 spores
- (E) 96 spores
- (F) 200 spores
- (G) 2000 spores

34. A transversal crosses two parallel lines. A pair of corresponding angles measure $(3x)$ degrees and $(x + 50)$ degrees. What is the value of x ? [[Lines & Angles](#)]

- (A) 100
- (B) 30
- (C) 12.5
- (D) 50
- (E) 75
- (F) 25
- (G) 20

35. In a village of 800 people, 35 percent fall ill because of poor sanitation. How many people stay healthy?

[\[Wastewater Story\]](#)

- (A) 765 people
- (B) 260 people
- (C) 480 people
- (D) 280 people
- (E) 520 people
- (F) 320 people
- (G) 535 people

36. A cycle priced at Rs 4000 has its price raised by 15 percent and then lowered by 15 percent. What is the final price? [\[Comparing Quantities\]](#)

- (A) Rs 4600
- (B) Rs 3400
- (C) Rs 4090
- (D) Rs 3850
- (E) Rs 3900
- (F) Rs 3910
- (G) Rs 4000

37. A seed weighs 0.8 grams, and its embryo makes up 15 percent of that mass. What is the mass of the embryo in milligrams? [\[Reproduction in Plants\]](#)

- (A) 12 mg
- (B) 120 mg
- (C) 0.12 mg
- (D) 800 mg
- (E) 1.2 mg
- (F) 68 mg

38. Two angles forming a linear pair are in the ratio 4 : 5. What is the measure of the smaller angle? [\[Lines & Angles\]](#)

- (A) 60 degrees
- (B) 80 degrees
- (C) 90 degrees
- (D) 72 degrees
- (E) 108 degrees
- (F) 100 degrees

39. A treatment plant collects 1.5 tonnes of sludge each day. Over 30 days, how many kilograms of sludge does it collect in total? [\[Wastewater Story\]](#)

- (A) 45 kg
- (B) 45000 kg
- (C) 30000 kg
- (D) 450000 kg
- (E) 4500 kg
- (F) 15000 kg
- (G) 1500 kg

40. In a class of 45 students, 60 percent are boys. How many girls are in the class? [\[Comparing Quantities\]](#)

- (A) 15 girls
- (B) 18 girls
- (C) 27 girls
- (D) 9 girls
- (E) 30 girls
- (F) 24 girls
- (G) 20 girls

41. In a garden, 40 percent of 350 flowers are cross-pollinated and the rest are self-pollinated. How many flowers are self-pollinated? [\[Reproduction in Plants\]](#)

- (A) 240 flowers
- (B) 35 flowers
- (C) 310 flowers
- (D) 60 flowers
- (E) 290 flowers
- (F) 210 flowers

42. Two supplementary angles differ from each other by 40 degrees. What is the measure of the smaller angle? [\[Lines & Angles\]](#)

- (A) 80 degrees
- (B) 90 degrees
- (C) 70 degrees
- (D) 50 degrees
- (E) 60 degrees
- (F) 110 degrees
- (G) 100 degrees

43. An old toilet flushes with 9 litres of water, while a new dual-flush toilet uses only 3 litres. What percentage of water is saved on each flush? [\[Wastewater Story\]](#)

- (A) 300%
- (B) 50%
- (C) 200%
- (D) 75%
- (E) 66.7%
- (F) 60%
- (G) 33.3%

44. A trader sells an item for Rs 575 at a loss of 8 percent. What was the cost price of the item? [\[Comparing Quantities\]](#)

- (A) Rs 621
- (B) Rs 500
- (C) Rs 529
- (D) Rs 600
- (E) Rs 690
- (F) Rs 650
- (G) Rs 625

45. Of 20 coconuts that float away on the sea, only 8 reach a new shore and sprout. What fraction of the coconuts fail to sprout, in simplest form? [\[Reproduction in Plants\]](#)

- (A) $\frac{1}{5}$
- (B) $\frac{1}{2}$
- (C) $\frac{3}{5}$
- (D) $\frac{3}{4}$
- (E) $\frac{8}{20}$
- (F) $\frac{12}{20}$
- (G) $\frac{2}{5}$

46. One angle measures 250 degrees. What is the measure of the smaller (non-reflex) angle formed on the other side of the same arms? [\[Lines & Angles\]](#)

- (A) 80 degrees
- (B) 70 degrees
- (C) 90 degrees
- (D) 250 degrees
- (E) 160 degrees
- (F) 130 degrees
- (G) 110 degrees

47. A treatment plant runs for 22 hours a day at a rate of 15000 litres per hour. How many kilolitres of water does it treat in one day? [\[Wastewater Story\]](#)

- (A) 220 kL
- (B) 300 kL
- (C) 330 kL
- (D) 360 kL
- (E) 33 kL
- (F) 150 kL

48. The ages of two children are in the ratio 2 : 3, and the younger child is 24 years old. How old is the older child? [\[Comparing Quantities\]](#)

- (A) 30 years
- (B) 16 years
- (C) 40 years
- (D) 48 years
- (E) 36 years
- (F) 32 years
- (G) 18 years

49. A flower's ovary holds 24 ovules, and after fertilisation 18 of them become seeds. What percentage of the ovules turned into seeds? [\[Reproduction in Plants\]](#)

- (A) 24%
- (B) 25%
- (C) 18%
- (D) 6%
- (E) 75%
- (F) 72%

- 50.** A transversal cuts two parallel lines, making alternate interior angles of $(5x - 20)$ degrees and $(3x + 40)$ degrees. What is the measure of each of these angles? [\[Lines & Angles\]](#)
- (A) 130 degrees
 - (B) 90 degrees
 - (C) 70 degrees
 - (D) 30 degrees
 - (E) 110 degrees
 - (F) 150 degrees
 - (G) 100 degrees
- 51.** Of 500 kilolitres of treated water, 40 percent is reused for gardens and the rest flows into a river. How many kilolitres flow into the river? [\[Wastewater Story\]](#)
- (A) 200 kL
 - (B) 250 kL
 - (C) 540 kL
 - (D) 460 kL
 - (E) 300 kL
 - (F) 100 kL
- 52.** A 250 millilitre drink is 12 percent sugar by volume, and 1 millilitre of sugar weighs about 1 gram. How many grams of sugar does the drink contain? [\[Comparing Quantities\]](#)
- (A) 25 g
 - (B) 3 g
 - (C) 15 g
 - (D) 12 g
 - (E) 30 g
 - (F) 300 g
 - (G) 36 g
- 53.** A strawberry runner grows a brand-new plant at every 30 centimetres along its length. How many new plants form along a runner that is 4.5 metres long? [\[Reproduction in Plants\]](#)
- (A) 16 plants
 - (B) 30 plants
 - (C) 1.5 plants
 - (D) 150 plants
 - (E) 15 plants
 - (F) 13.5 plants
- 54.** The complement of an angle is four times the angle itself. What is the measure of the angle? [\[Lines & Angles\]](#)
- (A) 72 degrees
 - (B) 45 degrees
 - (C) 30 degrees
 - (D) 36 degrees
 - (E) 22.5 degrees
 - (F) 18 degrees
 - (G) 15 degrees

- 55.** A family of 5 people each use 135 litres of water per day. How many kilolitres of water does the family use in a 30-day month? [\[Wastewater Story\]](#)
- (A) 20.25 kL
 - (B) 2.5 kL
 - (C) 675 kL
 - (D) 202.5 kL
 - (E) 4050 kL
 - (F) 2.025 kL
- 56.** An item is bought for Rs 750 and sold for Rs 900. What is the profit percentage? [\[Comparing Quantities\]](#)
- (A) 15%
 - (B) 25%
 - (C) 150%
 - (D) 20%
 - (E) 30%
 - (F) 18%
- 57.** A farmer sows 5 kilograms of wheat seed and harvests 60 kilograms of grain. The harvested grain is how many times the mass of the seed sown? [\[Reproduction in Plants\]](#)
- (A) 300 times
 - (B) 1.2 times
 - (C) 65 times
 - (D) 12 times
 - (E) 55 times
 - (F) 10 times
- 58.** Around a single point, three angles measure 90 degrees, 120 degrees and 85 degrees, with one more angle completing the turn. What is the fourth angle? [\[Lines & Angles\]](#)
- (A) 75 degrees
 - (B) 65 degrees
 - (C) 45 degrees
 - (D) 90 degrees
 - (E) 55 degrees
 - (F) 85 degrees
 - (G) 295 degrees
- 59.** Wastewater entering a plant is part domestic and part industrial in the ratio 8 : 3. What percentage of the total wastewater is industrial? [\[Wastewater Story\]](#)
- (A) 11%
 - (B) 3%
 - (C) 72.7%
 - (D) 27.3%
 - (E) 8%
 - (F) 30%

60. 35 percent of a certain number is 84. What is the number? [\[Comparing Quantities\]](#)

- (A) 29.4
- (B) 84
- (C) 240
- (D) 210
- (E) 300
- (F) 120

61. A bee pollinates 3 out of every 5 flowers it visits. If it visits 75 flowers, how many of them are NOT pollinated? [\[Reproduction in Plants\]](#)

- (A) 50 flowers
- (B) 60 flowers
- (C) 15 flowers
- (D) 45 flowers
- (E) 72 flowers
- (F) 30 flowers
- (G) 25 flowers

62. Two co-interior angles between parallel lines are such that one is twice the other. What is the measure of the larger angle? [\[Lines & Angles\]](#)

- (A) 72 degrees
- (B) 108 degrees
- (C) 120 degrees
- (D) 100 degrees
- (E) 90 degrees
- (F) 135 degrees
- (G) 60 degrees

63. Treatment lowers the pollution level (BOD) of water from 250 milligrams per litre down to 30 milligrams per litre. What is the percentage reduction in BOD? [\[Wastewater Story\]](#)

- (A) 12%
- (B) 220%
- (C) 85%
- (D) 8.8%
- (E) 30%
- (F) 80%
- (G) 88%

64. A 40-litre mixture contains water and milk in the ratio 3 : 5. How many litres of milk are in the mixture?

[\[Comparing Quantities\]](#)

- (A) 16 L
- (B) 20 L
- (C) 30 L
- (D) 15 L
- (E) 18 L
- (F) 24 L
- (G) 25 L

65. Seeds of a plant sprout in 6 days at 30 degrees Celsius, but take 50 percent longer at 20 degrees Celsius. How many days do they take at 20 degrees? [\[Reproduction in Plants\]](#)

- (A) 18 days
- (B) 3 days
- (C) 7.5 days
- (D) 6.5 days
- (E) 12 days
- (F) 8 days
- (G) 9 days

66. An angle and its supplement are in the ratio 1 : 5. What is the measure of the angle? [\[Lines & Angles\]](#)

- (A) 150 degrees
- (B) 15 degrees
- (C) 36 degrees
- (D) 45 degrees
- (E) 90 degrees
- (F) 60 degrees
- (G) 30 degrees

67. Manholes are placed every 30 metres along a sewer line that is 1.5 kilometres long, with one at each end. How many manholes are needed in all? [\[Wastewater Story\]](#)

- (A) 51 manholes
- (B) 50 manholes
- (C) 49 manholes
- (D) 30 manholes
- (E) 500 manholes
- (F) 52 manholes
- (G) 45 manholes

68. A phone's price falls from Rs 20000 to Rs 17000. What is the percentage decrease in price? [\[Comparing Quantities\]](#)

- (A) 20%
- (B) 12%
- (C) 17.6%
- (D) 85%
- (E) 15%
- (F) 3%
- (G) 30%

69. A mango tree bears 1500 flowers, but only 2 percent of them grow into fruits. How many mangoes does the tree produce? [\[Reproduction in Plants\]](#)

- (A) 300 mangoes
- (B) 30 mangoes
- (C) 750 mangoes
- (D) 75 mangoes
- (E) 20 mangoes
- (F) 150 mangoes
- (G) 3 mangoes

70. Two angles forming a linear pair measure $(3x + 15)$ degrees and $(2x + 25)$ degrees. What is the measure of the obtuse angle? [\[Lines & Angles\]](#)

- (A) 105 degrees
- (B) 90 degrees
- (C) 84 degrees
- (D) 81 degrees
- (E) 100 degrees
- (F) 99 degrees
- (G) 96 degrees

71. A bucket bath uses 18 litres of water. A shower runs at 9 litres per minute for 6 minutes. Which uses more water, and by how much? [\[Wastewater Story\]](#)

- (A) Bucket by 9 L
- (B) Equal
- (C) Bucket by 36 L
- (D) Shower by 36 L
- (E) Shower by 18 L
- (F) Shower by 54 L

72. A man spends 70 percent of his monthly income and saves the rest, which comes to Rs 4500. What is his monthly income? [\[Comparing Quantities\]](#)

- (A) Rs 15000
- (B) Rs 6428
- (C) Rs 4500
- (D) Rs 10500
- (E) Rs 3150
- (F) Rs 9000
- (G) Rs 150000

73. Wind carries one plant's seeds about 60 metres, while an animal carries another plant's seeds about 15 metres. What is the ratio of the wind distance to the animal distance, in simplest form? [\[Reproduction in Plants\]](#)

- (A) 1 : 4
- (B) 60 : 15
- (C) 3 : 1
- (D) 5 : 1
- (E) 4 : 3
- (F) 4 : 1
- (G) 45 : 1

74. Two complementary angles are in the ratio 7 : 8. What is the measure of the larger angle? [\[Lines & Angles\]](#)

- (A) 54 degrees
- (B) 40 degrees
- (C) 48 degrees
- (D) 60 degrees
- (E) 45 degrees
- (F) 56 degrees
- (G) 42 degrees

75. A test shows that every 250 millilitres of dirty water carries 5 harmful germs. How many such germs are present in 2 litres of the same water? [\[Wastewater Story\]](#)

- (A) 80 germs
- (B) 40 germs
- (C) 10 germs
- (D) 400 germs
- (E) 20 germs
- (F) 5 germs

76. An article priced at Rs 2000 is given two successive discounts of 20 percent and then 10 percent. This equals a single discount of what percentage? [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 70%
- (C) 32%
- (D) 72%
- (E) 28%
- (F) 30%

77. A certain flower opens at 6 a.m. and closes at 6 p.m. every day. Over a full 4-week month of 28 days, for how many hours in total is the flower open? [\[Reproduction in Plants\]](#)

- (A) 288 hours
- (B) 336 hours
- (C) 168 hours
- (D) 144 hours
- (E) 360 hours
- (F) 84 hours

78. On a clock face, the hands point at 12 and at 4. What is the smaller angle between the two hands? [\[Lines & Angles\]](#)

- (A) 90 degrees
- (B) 150 degrees
- (C) 100 degrees
- (D) 120 degrees
- (E) 60 degrees
- (F) 160 degrees

79. Cleaning 1000 litres of wastewater costs Rs 6. What is the cost of treating 360000 litres of wastewater? [\[Wastewater Story\]](#)

- (A) Rs 216
- (B) Rs 360
- (C) Rs 600
- (D) Rs 3600
- (E) Rs 1080
- (F) Rs 2160

80. A sum of money grows to Rs 6600 in 2 years under simple interest at 10 percent per year. What was the principal? [\[Comparing Quantities\]](#)

- (A) Rs 5280
- (B) Rs 5800
- (C) Rs 6000
- (D) Rs 5000
- (E) Rs 6300
- (F) Rs 5400
- (G) Rs 5500

81. A plant bears 80 flowers, of which 25 percent are unisexual and the rest are bisexual. How many bisexual flowers does it have? [\[Reproduction in Plants\]](#)

- (A) 65 flowers
- (B) 60 flowers
- (C) 30 flowers
- (D) 20 flowers
- (E) 55 flowers
- (F) 40 flowers
- (G) 75 flowers

82. Two supplementary angles are such that one is two-thirds of the other. What is the measure of the smaller angle? [\[Lines & Angles\]](#)

- (A) 80 degrees
- (B) 90 degrees
- (C) 108 degrees
- (D) 72 degrees
- (E) 60 degrees
- (F) 100 degrees
- (G) 120 degrees

83. Algae on a polluted pond double in area every 2 days. After 6 days, the algae cover is how many times the area it started with? [\[Wastewater Story\]](#)

- (A) 16 times
- (B) 12 times
- (C) 9 times
- (D) 8 times
- (E) 4 times
- (F) 6 times
- (G) 3 times

84. A's salary is 25 percent more than B's salary. By what percentage is B's salary less than A's? [\[Comparing Quantities\]](#)

- (A) 15%
- (B) 80%
- (C) 20%
- (D) 75%
- (E) 33.3%
- (F) 25%

85. Each potato plant yields 1.2 kilograms of potatoes, but in a field of 250 plants, 8 percent of the plants die before harvest. What is the total harvest from the surviving plants? [\[Reproduction in Plants\]](#)

- (A) 312 kg
- (B) 276 kg
- (C) 240 kg
- (D) 264 kg
- (E) 220.8 kg
- (F) 27.6 kg

86. Two angles lie next to each other on a straight line and measure $(x + 30)$ degrees and $(2x - 15)$ degrees. What is the measure of the larger angle? [\[Lines & Angles\]](#)

- (A) 75 degrees
- (B) 110 degrees
- (C) 85 degrees
- (D) 90 degrees
- (E) 100 degrees
- (F) 80 degrees
- (G) 95 degrees

87. A septic tank fills with 250 litres of waste each day and has a capacity of 7500 litres. In how many days will it be full? [\[Wastewater Story\]](#)

- (A) 300 days
- (B) 15 days
- (C) 20 days
- (D) 3 days
- (E) 30 days
- (F) 25 days
- (G) 35 days

88. A box of 24 pens costs Rs 144, and the pens are sold at Rs 8 each. What is the profit percentage?

[\[Comparing Quantities\]](#)

- (A) 25%
- (B) 8%
- (C) 50%
- (D) 20%
- (E) 30%
- (F) 16.7%
- (G) 33.3%

89. A fern produces 8000 spores while a flowering plant produces only 50 seeds. The fern's spore count is how many times the flowering plant's seed count? [\[Reproduction in Plants\]](#)

- (A) 7950 times
- (B) 16 times
- (C) 60 times
- (D) 160 times
- (E) 1600 times
- (F) 100 times

- 90.** The supplement of an angle is 78 degrees more than twice its complement. What is the measure of the angle? [\[Lines & Angles\]](#)
- (A) 78 degrees
 - (B) 66 degrees
 - (C) 102 degrees
 - (D) 90 degrees
 - (E) 12 degrees
 - (F) 96 degrees
 - (G) 84 degrees
- 91.** A city of 2 lakh (200000) people uses 150 litres of water per person each day. How many million litres does the whole city use daily? [\[Wastewater Story\]](#)
- (A) 15 million L
 - (B) 300 million L
 - (C) 300000 million L
 - (D) 20 million L
 - (E) 30 million L
 - (F) 3 million L
 - (G) 60 million L
- 92.** Three-fifths of a sum of money is Rs 1800. What is the whole sum? [\[Comparing Quantities\]](#)
- (A) Rs 1500
 - (B) Rs 2700
 - (C) Rs 2400
 - (D) Rs 1080
 - (E) Rs 3000
 - (F) Rs 5000
 - (G) Rs 3600
- 93.** In a germination test, 9 of 12 seeds sprout in dish A, while 16 of 20 seeds sprout in dish B. Which dish has the higher germination rate, and by how many percentage points? [\[Reproduction in Plants\]](#)
- (A) B by 4 points
 - (B) A by 5 points
 - (C) B by 5 points
 - (D) B by 8 points
 - (E) B by 25 points
 - (F) Equal rates
 - (G) A by 3 points
- 94.** On a straight line, three adjacent angles are in the ratio 1 : 2 : 3. What is the measure of the largest angle? [\[Lines & Angles\]](#)
- (A) 100 degrees
 - (B) 30 degrees
 - (C) 45 degrees
 - (D) 90 degrees
 - (E) 60 degrees
 - (F) 80 degrees

- 95.** A hotel reuses 45 percent of the 8000 litres of wastewater it makes each day. How many litres does it reuse over a full 7-day week? [\[Wastewater Story\]](#)
- (A) 2520 L
 - (B) 3500 L
 - (C) 3600 L
 - (D) 25200 L
 - (E) 30800 L
 - (F) 36000 L
- 96.** The population of a town is 8000 and grows by 5 percent each year. What will the population be after 2 years? [\[Comparing Quantities\]](#)
- (A) 8000
 - (B) 8820
 - (C) 8400
 - (D) 8410
 - (E) 8800
 - (F) 8200
 - (G) 9261
- 97.** A pollen tube grows 0.5 millimetres every minute and must travel 6 millimetres to reach the ovule. How many minutes does it take? [\[Reproduction in Plants\]](#)
- (A) 6 minutes
 - (B) 30 minutes
 - (C) 24 minutes
 - (D) 3 minutes
 - (E) 12 minutes
 - (F) 5.5 minutes
 - (G) 11 minutes
- 98.** Two angles form a linear pair in which one angle is four times the other. What is the measure of the larger angle? [\[Lines & Angles\]](#)
- (A) 90 degrees
 - (B) 45 degrees
 - (C) 36 degrees
 - (D) 144 degrees
 - (E) 120 degrees
 - (F) 72 degrees
 - (G) 108 degrees
- 99.** At a plant, every 100 kilograms of sludge produces 6 cubic metres of biogas. How much biogas is produced from 2.5 tonnes of sludge? [\[Wastewater Story\]](#)
- (A) 75 cubic metres
 - (B) 15000 cubic metres
 - (C) 15 cubic metres
 - (D) 1500 cubic metres
 - (E) 250 cubic metres
 - (F) 150 cubic metres

100. A merchant marks his goods 40 percent above the cost price and then allows a 25 percent discount on the marked price. What is his profit percentage? [\[Comparing Quantities\]](#)

- (A) 15%
- (B) 10%
- (C) 40%
- (D) 5%
- (E) 20%
- (F) 25%
- (G) 4%